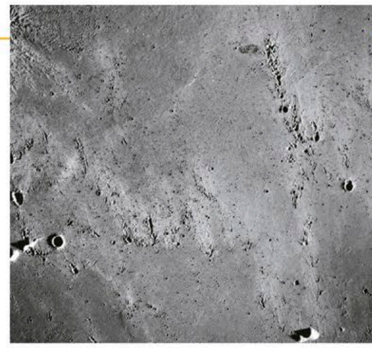
NEAR SIDE *northern hemisphere*

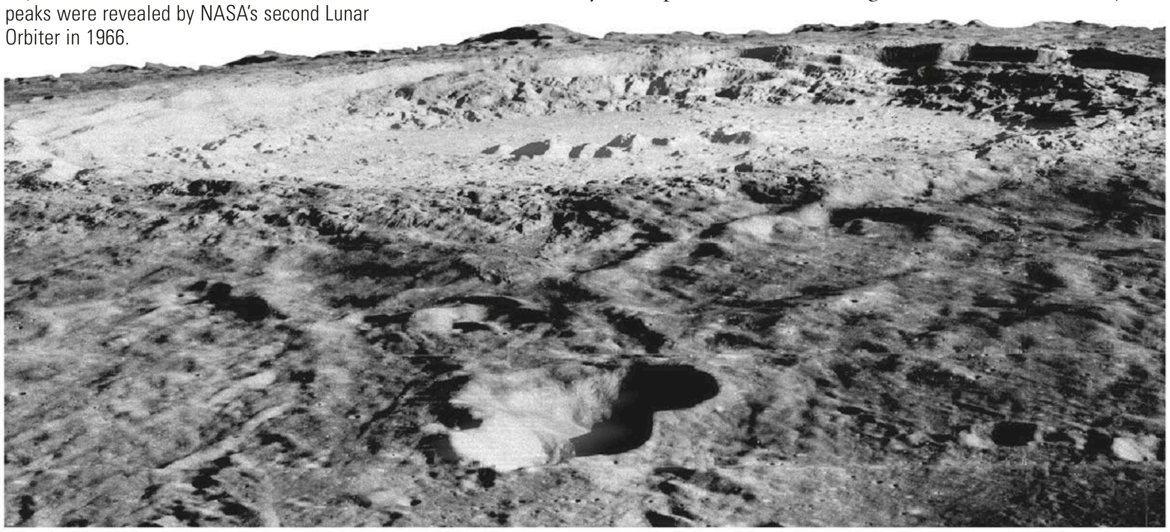
Copernicus Crater

TYPE Impact crater
AGE 900 million years
DIAMETER 57 miles (91 km)

This young ray crater has massive terraced walls. The crater floor is below the general level of the surrounding plain and lies 2.3 miles (3.7 km) below the top of the surrounding walls. Copernicus is an intermediate-sized crater with high central peaks. These mountains were formed when the rock directly below the crater rebounded after being compressed by the explosion caused by the impacting asteroid. The vicinity of Copernicus is peppered with secondary craters formed by boulders thrown out during the impact. Fine, light gray rock particles ejected during the crater's formation were collected by the Apollo 12 astronauts near their landing site. Such particles were responsible for forming the rays that surround the crater. The high reflectivity of the rays is due to the ejecta churning up the lunar regolith. (Rough material reflects more light than smooth material.)

CRATER CHAINS
The material excavated by an impact showers down on the surrounding lunar surface, producing long chains of secondary craters.





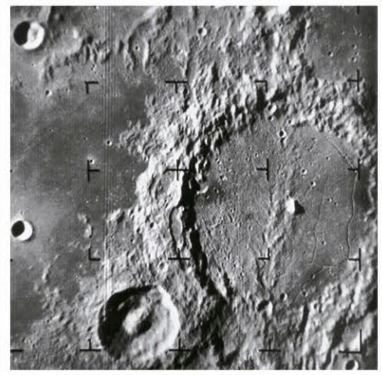
LUNAR ORBITER 2 IMAGE
Copernicus Crater's terraced walls and central peaks were revealed by NASA's second Lunar Orbiter in 1966.

NEAR SIDE *southern hemisphere*

Alphonsus Crater

TYPE Impact crater
AGE 4.0 billion years
DIAMETER 80 miles (117 km)

NASA's Ranger 9 spacecraft was deliberately crash-landed into the Alphonsus Crater on March 24, 1965, taking television pictures as it approached. The crater formed in an impact, but the dark patches and fractures Ranger 9 found on its floor are thought to be a result of volcanic activity—probably explosive eruptions. Because of these features, Alphonsus was considered a possible landing site for later Apollo missions.



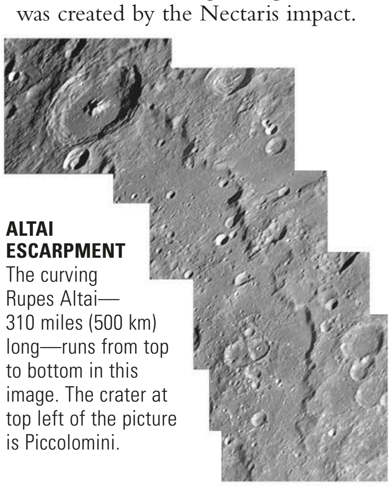
THREE MINUTES BEFORE IMPACT

NEAR SIDE *southern hemisphere*

Rupes Altai

TYPE Cliff
AGE 4.2 billion years
LENGTH 315 miles (507 km)

Altai is by far the longest cliff on the Moon. It is about 1.1 miles (1.8 km) high. The energy that is released during an impact does more than just excavate a crater and lift material out to form walls and an ejecta blanket. Violent seismic shock waves radiate away from the impact point. An obstacle such as a mountain can halt these waves and the lunar crust may then buckle, forming a long cliff. Altai was created by the Nectaris impact.



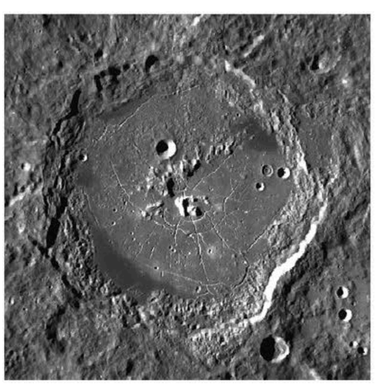
ALTAI ESCARPMENT
The curving Rupes Altai—310 miles (500 km) long—runs from top to bottom in this image. The crater at top left of the picture is Piccolomini.

NEAR SIDE *southern hemisphere*

Humboldt Crater

TYPE Impact crater
AGE About 3.8 billion years
DIAMETER 120 miles (189 km)

This crater is remarkable because its lava-filled floor is crisscrossed with a series of radial and concentric fractures (or rilles). On closer inspection, some look like collapsed tubes through which lava once flowed, and others like rift valleys. Lunar volcanic activity lasted for over 500 million years. Lava would seep up into a crater and then cool, shrink, crack, and sink. It would then be covered by more lava. The final basaltic infill would have many layers.



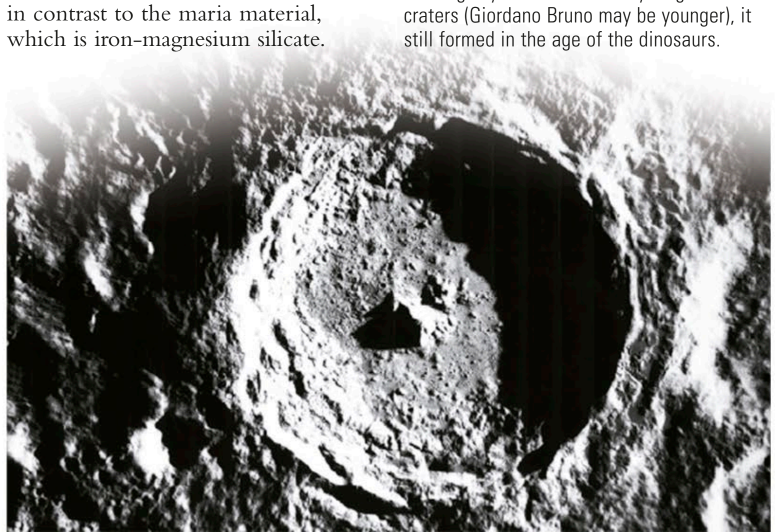
CRACKS ON THE FLOOR OF HUMBOLDT

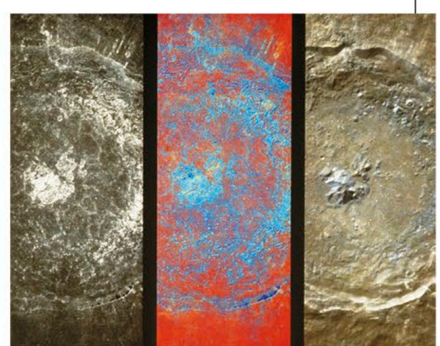
NEAR SIDE *southern hemisphere*

Tycho Crater

TYPE Impact crater
AGE 100 million years
DIAMETER 52 miles (85 km)

Lying in the southern highlands, Tycho is one of the most perfect walled craters on the Moon, with a central mountain peak towering 1.8 miles (3 km) above a rough infilled inner region. Surveyor 7 landed on the north rim of Tycho's ejecta blanket in January 1968. About 21,000 photographs were taken, and the soil was chemically analyzed. The highland soil was found to be mainly made of calcium-aluminum silicates, in contrast to the maria material, which is iron-magnesium silicate.





THREE FILTERED IMAGES OF TYCHO
The Ultraviolet/Visual camera onboard the Clementine spacecraft was equipped with a series of filters. Differing color combinations revealed the variability in the physical and chemical structure of the crater rock.

YOUNGEST LARGE LUNAR CRATER?
Although Tycho is one of the youngest lunar craters (Giordano Bruno may be younger), it still formed in the age of the dinosaurs.